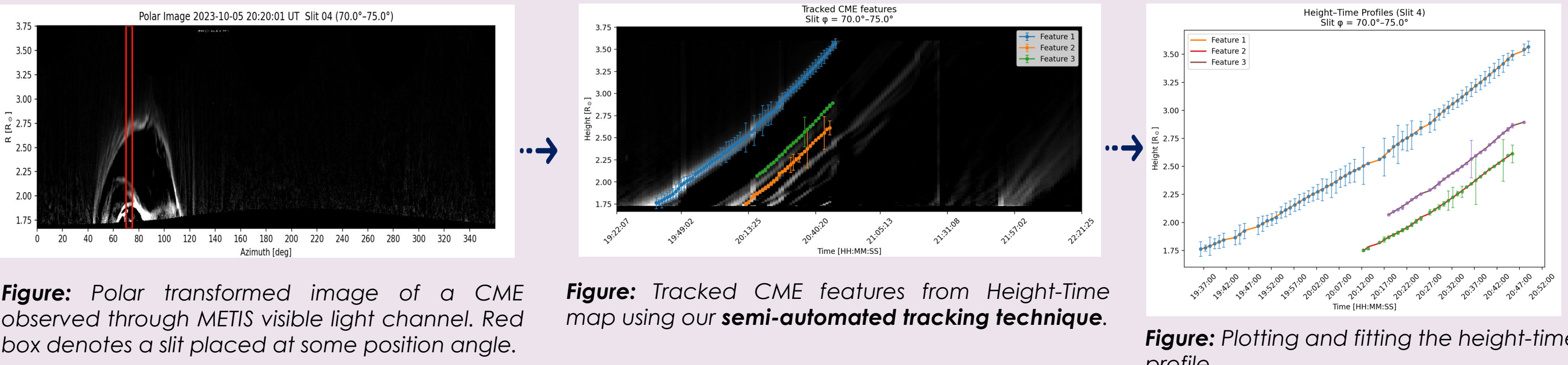


Coronal Mass Ejections (CMEs) are large-scale magnetized plasma eruptions driving severe space weather disturbances. Understanding their kinematics in the middle corona is essential for linking CME initiation to interplanetary evolution. Traditional manual tracking is subjective and often obscures velocity dispersion within components such as the core and front. To address this limitation, we formulated **DOFCAT** (*Dense Optical Flow CME Analysis Tool*), an **optical flow (OF)**-based technique to derive POS velocity fields from multi-instrument coronagraph observations (**LASCO C2**, **COR1/STEREO A**, **METIS/Solo**, and **PROBA3/ASPIICS**) spanning diverse coronal heights. Using robust processing to enhance CME-to-background contrast, the OF analysis captures spatiotemporal intensity variations across the structure, revealing **internal velocity dispersion**. Our results demonstrate that velocity profiles for the CME front and core are non-uniform and vary with position angle. In **impulsive CMEs**, the height offset between the front and core increases rapidly following the impulsive phase, specifically between **2.3-3.0 R_o**, whereas non-impulsive CMEs evolve more gradually. Furthermore, the CME front exhibits a higher velocity spread than the core. Ultimately, this work aims toward an **automated CME detection** and characterization pipeline using OF velocities for space-weather forecasting.

Abstract

Orthodox Methods for CME Kinematics



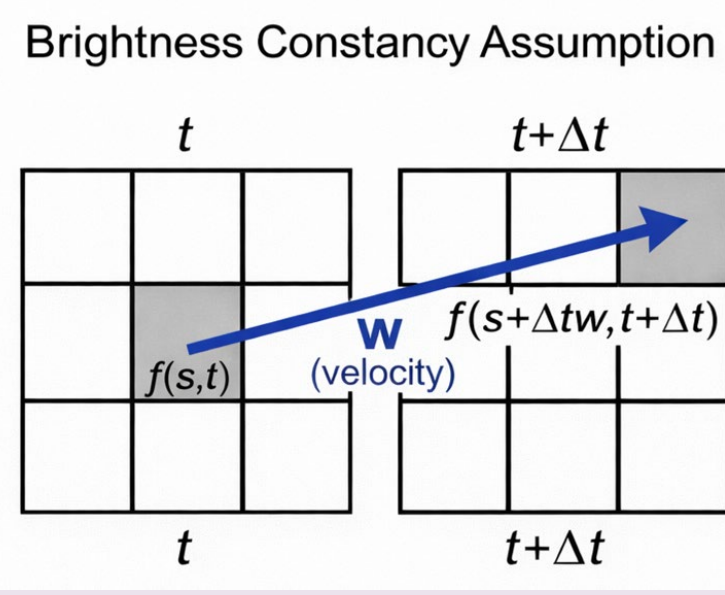
➤ Methods like this are not only time taking but also lead to subjectivity in measurements. Moreover, they are limited to measuring the velocity at only limited position angles and CME features.

What is Optical Flow?

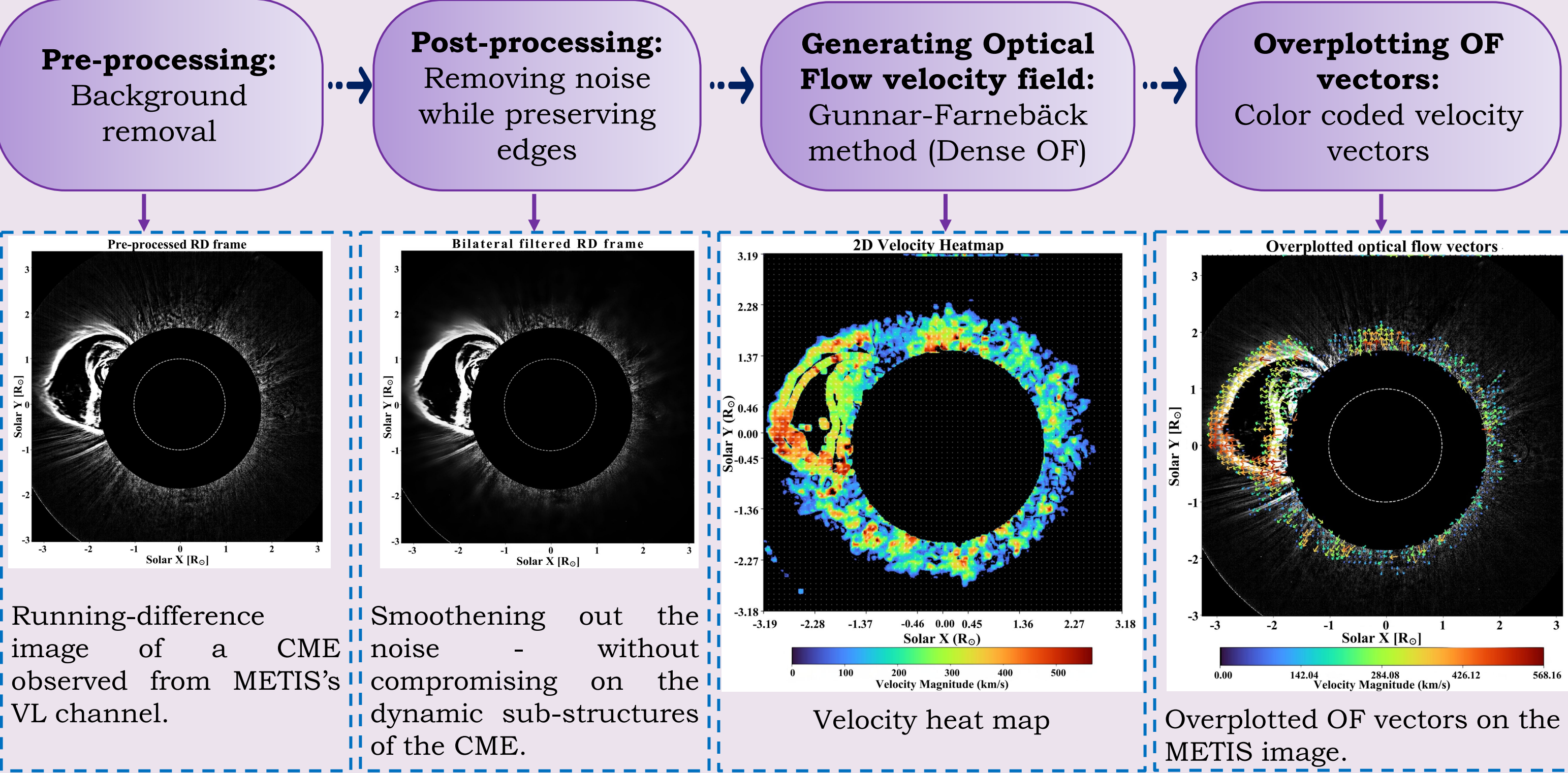
➤ Optical flow (OF) estimates motion by tracking intensity pattern shifts between consecutive images separated by a time interval Δt . The resulting two-dimensional vector field represents projected velocities of evolving coronal structures, providing a direct view of CME dynamics across the entire field of view. The method assumes that intensity remains approximately constant as structures move:

$$f(s + \Delta w, t + \Delta t) \approx f(s, t)$$

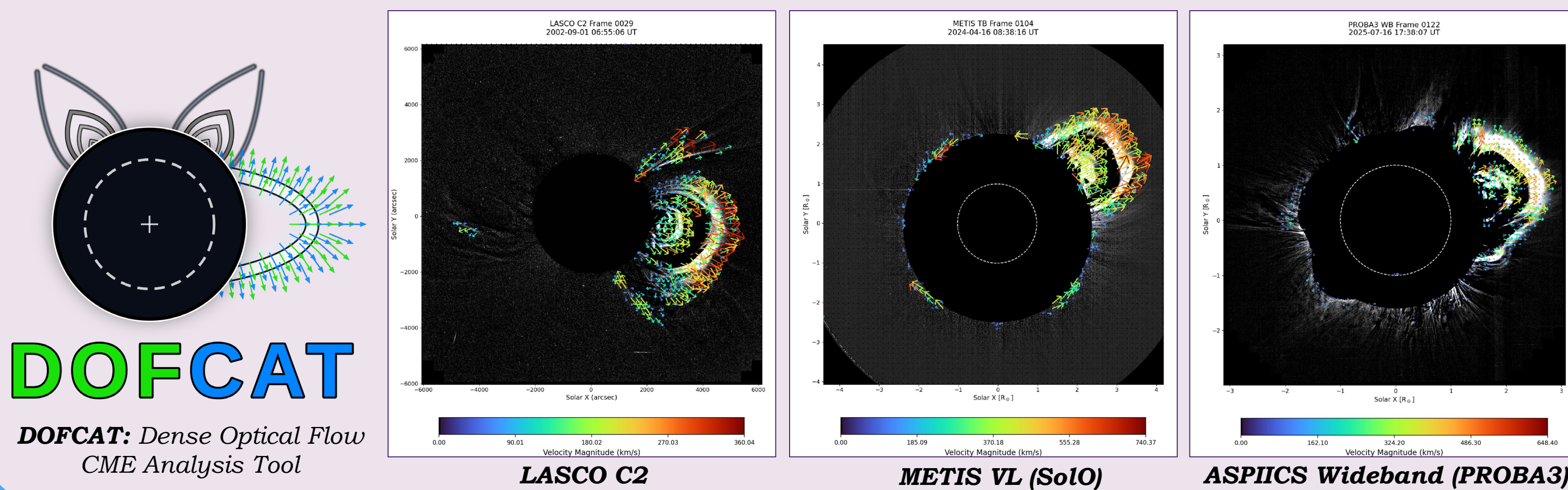
➤ Solving this equation produces dense velocity maps, enabling us to capture internal CME velocity dispersion- beyond what feature tracking can provide.



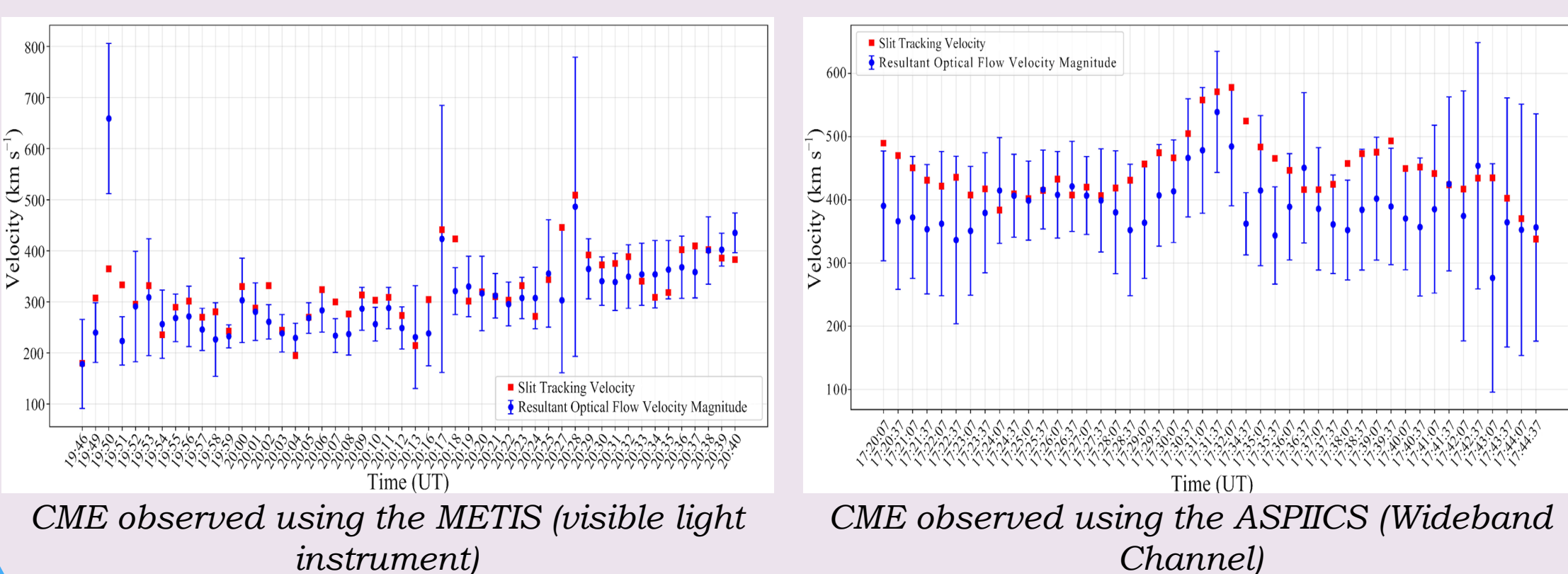
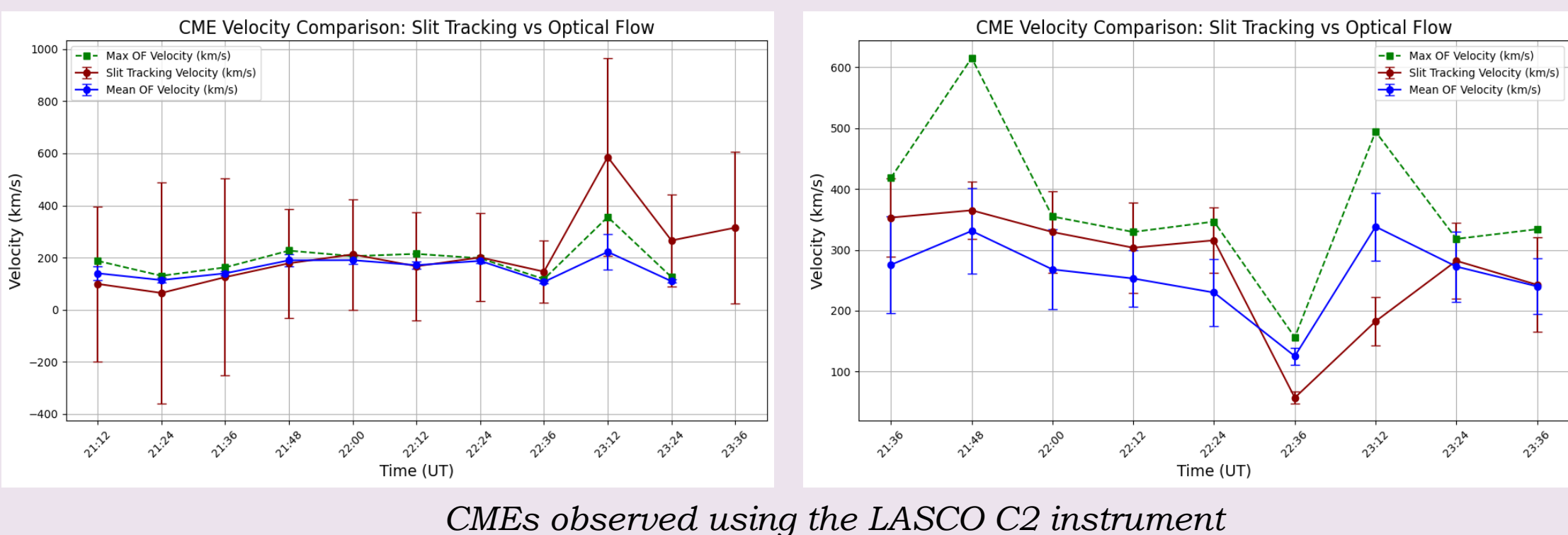
Methodology



Optical Flow on Multi-Coronagraph Data



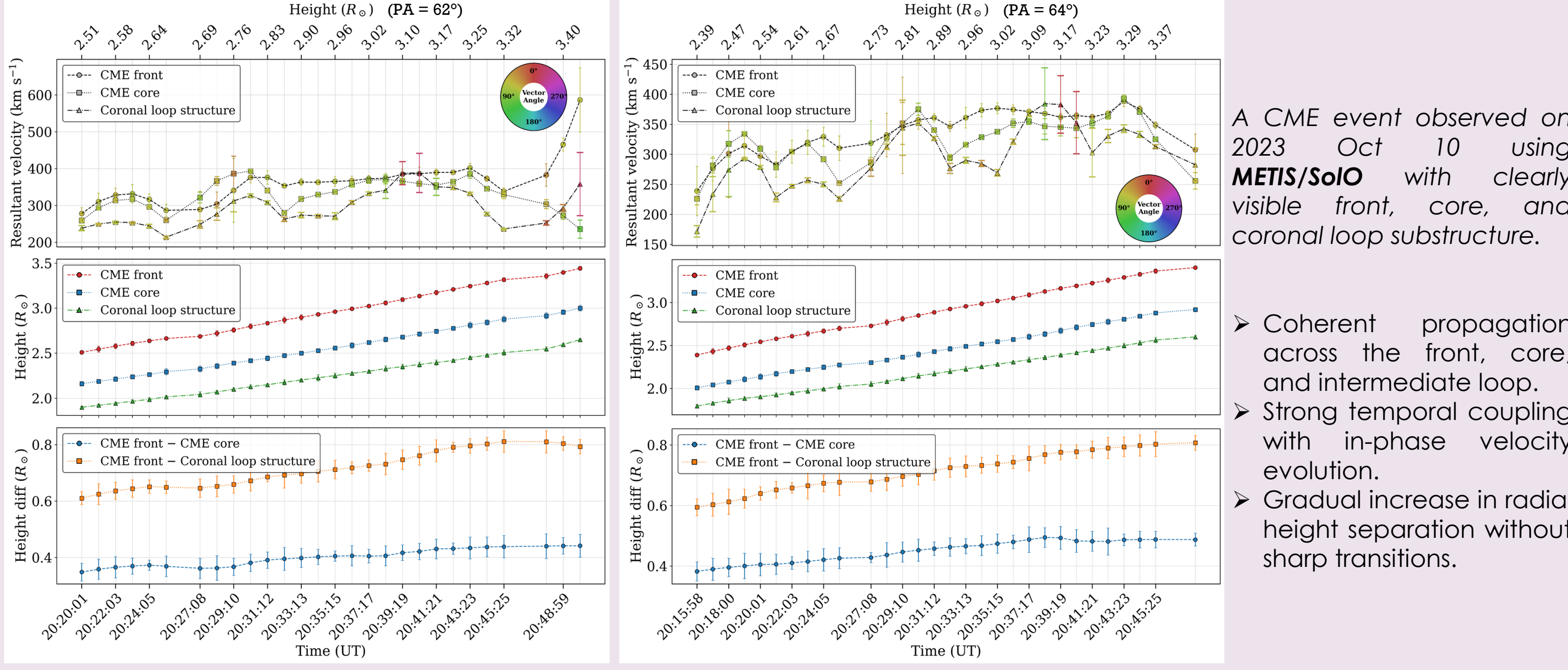
Validating DOFCAT



➤ The velocity-time profiles of the Optical Flow method matches quite well with the XT map tracking method and show similar temporal trends.

CME Velocity Dispersion and Analysing the Velocity Field

Figure(s): Velocity dispersion analysis between the CME front and other sub-structures at different position angles.

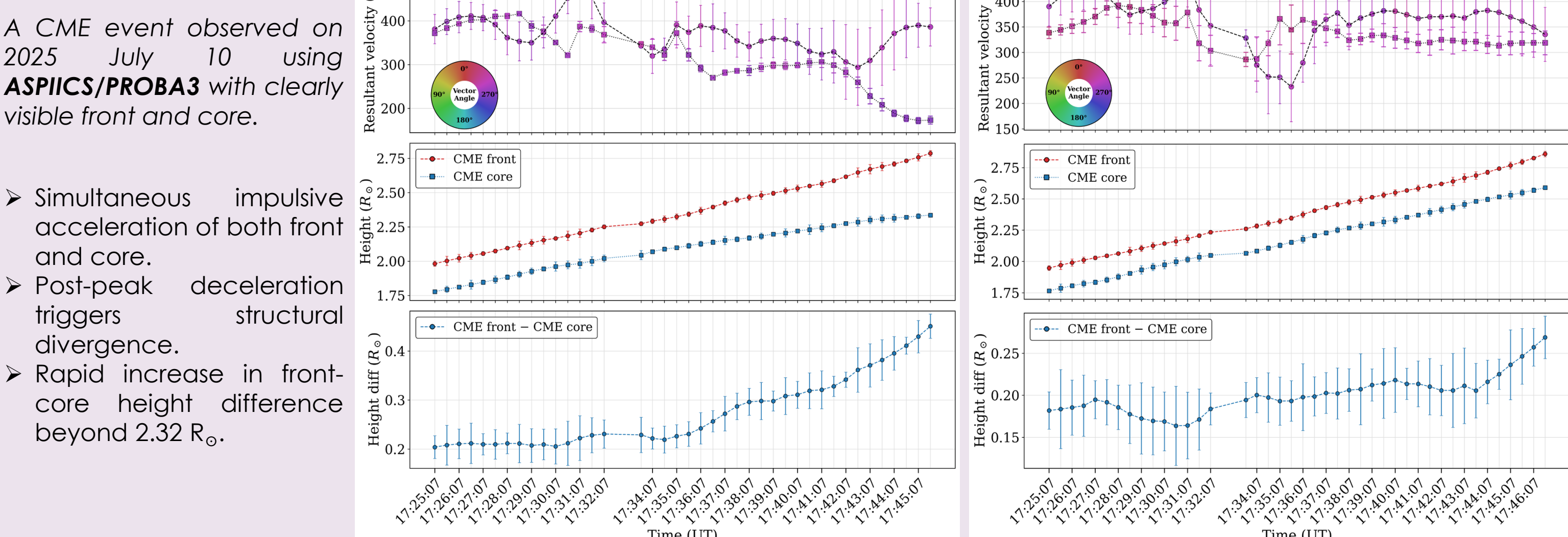


A CME event observed on 2023 Oct 10 using METIS/Solo with clearly visible front, core, and coronal loop substructure.

➤ Coherent propagation across the front, core, and intermediate loop.

➤ Strong temporal coupling with in-phase velocity evolution.

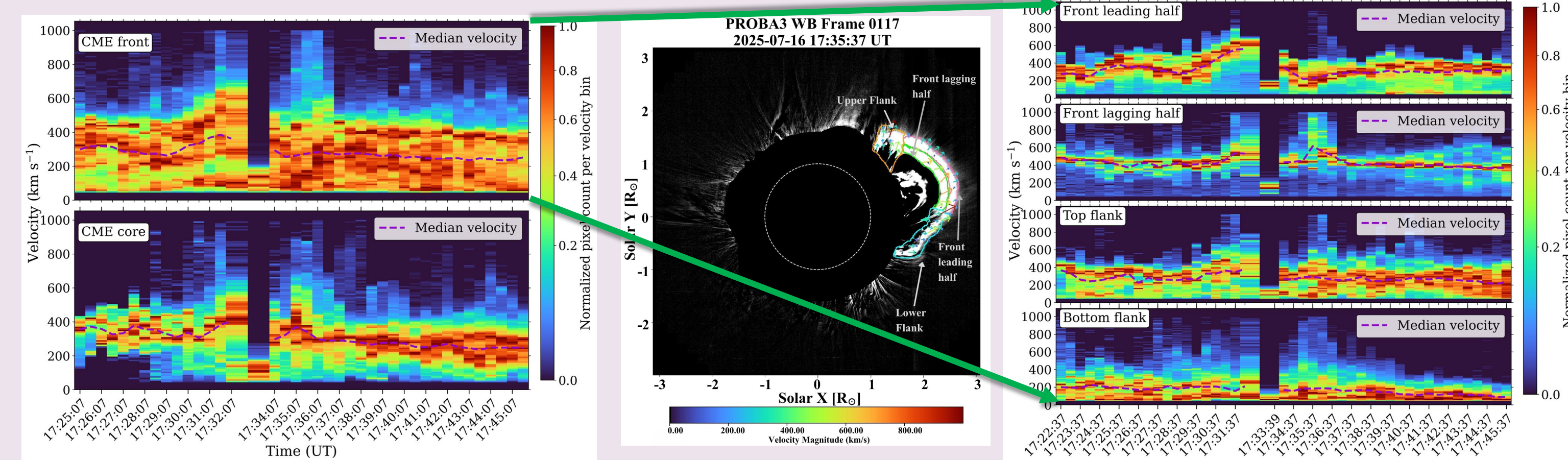
➤ Gradual increase in radial height separation without sharp transitions.



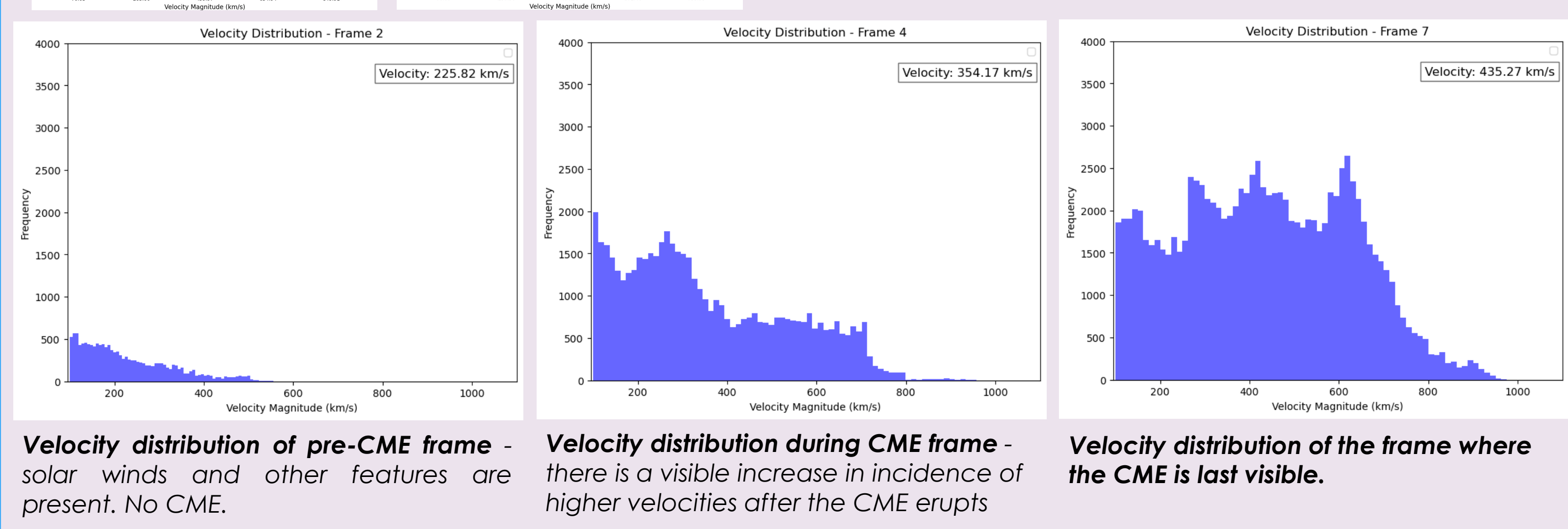
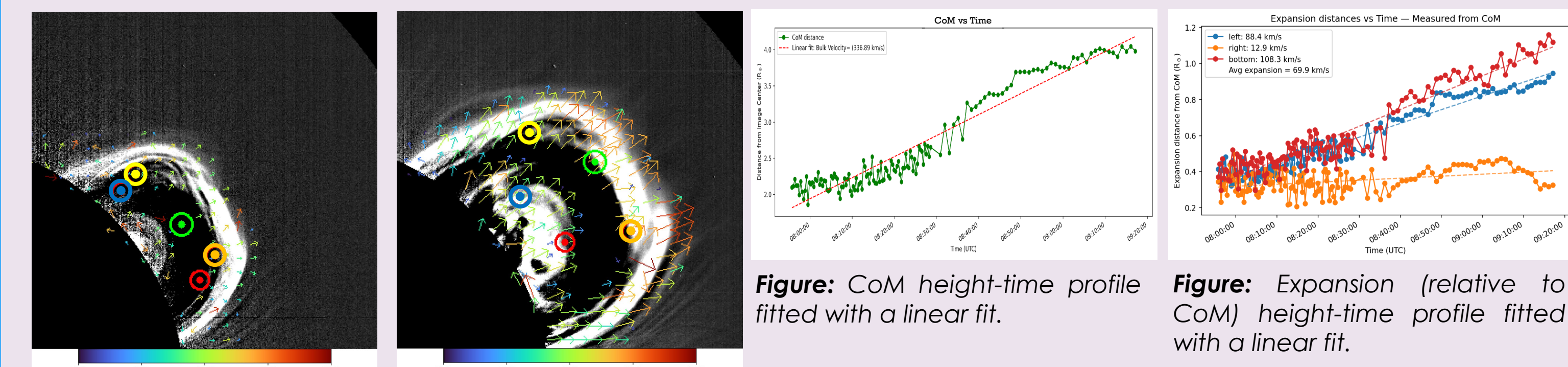
➤ Simultaneous impulsive acceleration of both front and core.

➤ Post-peak deceleration triggers structural divergence.

➤ Rapid increase in front-core height difference beyond 2.32 R_o.



Towards Automated CME detection



As the CME erupts, there is a notable increase in the incidence of higher velocities, as evident in the time evolution of the Histogram distribution. Based on this concept, we are training a ML model to detect CMEs, which is different from traditional image-based detection algorithms.

Results and Summary

- DOFCAT captures CME internal velocity dispersion, which is direction dependent.
- CME fronts evolve more heterogeneously than cores.
- Impulsive CMEs show rapid post-acceleration structural divergence.
- Front-Core Height dispersion starts at 2.3-3.0 R_o.
- Front-core S_y asymmetry explained by the front being squeezed between the core's outward push and ambient ram pressure, while the core is shielded.
- Changing velocity distributions will enable physics-based automated CME detection.